

EDUCATION

- **The University of Texas Austin** Austin, TX, USA
PhD Petroleum Engineering Aug. 2025 - Present
Courses: Subsurface ML, HPC for Engineers, Advanced Petrophysics
- **George Washington University** Washington D.C., USA
MSc in Computer Science; GPA: 3.9 Aug. 2023 - May 2025
Courses: Artificial Intelligence, Cloud Computing, Design & Analysis of Algorithms
Thesis: Regression on Interatomic Descriptor Data: Direct Solution Strategies for Linear Regression in CPU and Memory-Constrained Environments
- **Baku Higher Oil School** Baku, Azerbaijan
BSc Petroleum Engineering; GPA: 3.4 Sep. 2018 - June 2023
Courses: Geoscience, Formation Evaluation, Reservoir Flow Simulation
Thesis: Well Placement and Control Optimization with Numerical Optimization Algorithms

EXPERIENCE

- **DIRECT Consortium at UT Austin** Aug. 2025 – Present
Graduate Research Assistant
 - **Bridging** subsurface data and artificial intelligence to develop data-driven solutions for reservoir modeling and management under the supervision of Prof. Michael Pyrcz (PGE) and Prof. John Foster (Oden Institute).
 - **Developing** Physics-Informed Neural Networks (PINNs) and GANs to model subsurface flow by enforcing physical consistency within deep learning architectures.
- **BP** Baku, Azerbaijan
Reservoir Engineer / Data Analyst Sep. 2023 - Aug. 2025
 - **Analyzed** extensive surveillance data from the biggest oil field in the region.
 - **Saved** 300 man-hours per year using Python and SQL scripts.
 - **Integrated** GAP and Python to analyze the impact of pressure differentials on the vertical allocation of fluids, enhancing the accuracy of production system modeling.
 - **Developed** a statistical Python tool to post-process ensemble output and perform Monte Carlo simulations for production profiles. The tool provides sensitivity controls for project timings and well delivery schedules, saving a considerable amount of run and post-processing time.
 - **Post-processed** reservoir simulation outputs on HPC servers for automated sensitivity analysis and interpretation of communication between production and injection wells.
 - **Mentored** an intern in the development of an ML model to estimate a correction factor that adjusts relative permeability predictions in three-phase flow, significantly enhancing the accuracy.
 - **Ambassador** for Generative AI adoption and integration into workflows
- Reservoir Engineer Intern* June 2022 - Feb. 2023
 - **Analyzed** tracer data to assess sweep efficiency, influencing drilling decisions valued at \$4M.
 - **Estimated** surfactant flood opportunities for a mature brownfield, identifying potential for recovery through advanced chemical EOR techniques.
- **ei-Link Research and Development Center** Baku, Azerbaijan
Research Assistant Mar. 2021 - June 2022
 - **Comprehensive Literature Review** of expert systems and pattern recognition algorithms.
 - **Architected and developed** the backend for a web application toolkit featuring extensive reservoir engineering calculations, visualization capabilities, and a built-in data mining module.
 - **Boosted performance** of the app by 220% with integration of multithreading and caching mechanisms.

RESEARCH

- **Dashdamirov D.**, "Comparative Analysis: Violence Recognition from Videos using Computer Vision" 2024 IEEE 18th International Conference on Application of Information and Communication Technologies (AICT), Turin, Italy, 2024, pp. 1-6, doi: 10.1109/AICT61888.2024.10740414.

TEACHING EXPERIENCE

- **Machine Learning Instructor.** Lectured on Statistics, Feature Engineering, and ML algorithms in collaboration with non-profit organizations such as BHOS SPE Student Chapter and Youth Organization, *Feb. 2023 - May 2023*.
- **Academic Content Creator.** Translated and edited Physics content at Khan Academy to support free and accessible education, *Jul. 2020 - Jan. 2021*.

PROJECTS

- **SlugGuard – Expert System for AI-Driven Slugging Detection in Wells** A framework for detecting slugging in wells. The framework includes the end-to-end architecture of the system, including mimicking live sensor data, a database to store the features and labels, a rule-based decision agent, and a notification system.
Tech: Python, Experta, SQL
- **Overfit Potential of ANNs and its diagnosis** Demonstrated the susceptibility of neural networks to overfitting and designed multiple diagnostic and model-checking strategies to evaluate true generalization beyond conventional validation metrics.
Tech: Python, PyTorch, TensorFlow, OpenCV
- **ePattern** A web-based customizable visualization tool developed at ei-Link R&D. The app enables engineers to analyze data beyond standard visualization, evaluate reservoir health across areas, and perform detailed diagnostics of key parameters.
Tech: C#, SQL, Angular, REST APIs, Statistics
- **Loan Default Prediction with ML** Part of ABB Innovation Center - Machine Learning BootCamp. Predicted probability of bank customers defaulting on loans using advanced ML techniques and secured 1st place in the competition.
Tech: Python, Scikit-learn, XGBoost, LightGBM

HONORS AND AWARDS

- Awarded a fellowship by the Cockrell School of Engineering and the UT Austin Graduate School in recognition of academic achievements and research potential.
- Awarded Government Scholarship for dual degree MSc in Computer Science and Data Analytics at GW
- Winner of “Play to win 2024” award at BP for innovative projects and adding efficiency to workflows
- Graduated with honors from a Machine Learning traineeship program offered by ABB Innovation Center
- Received a Government Scholarship in recognition of academic achievement during 2019–2023
- Ranked in the top 3% of Azerbaijani university entrants based on admission exam scores

SKILLS

- **Languages:** Python, C#, R, Julia, SQL, Bash
- **Frameworks:** TensorFlow, PyTorch, Scikit-learn, OpenCV, Django
- **Tools & Platforms:** Eclipse, OPM, IPM Suite, Spotfire, Power BI, Docker, Git
- **Soft Skills:** Active listening, decision-making, teamwork, agility